

TOULOUSE BLAGNAC AP, France

WMO: 076300

Lat: 43.6211N

Lon: 1.3789E

Elev: 152

StdP: 99.51

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -3.7	(c) -2.0	(d) -7.1	(e) 2.1	(f) -0.6	(g) -5.3	(h) 2.5	(i) 0.3	(j) 12.3	(k) 10.4	(l) 11.0	(m) 9.5	(n) 2.4	(o) 280	(p) 0.510

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.2	33.3	20.8	31.5	20.3	29.8	19.8	22.3	29.9	21.6	28.8	20.8	27.7	4.0	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
20.0	15.0	25.3	19.1	14.2	24.7	18.4	13.5	24.0	66.1	29.8	63.2	29.0	60.6	27.7	29.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 10.8	(b) 9.4	(c) 8.4	(e) -5.8	(f) 36.4	(g) 2.1	(h) 2.1	(i) -7.3	(j) 37.9	(k) -8.5	(l) 39.1	(m) -9.7	(n) 40.2	(o) -11.2	(p) 41.7		
			DB	WB	DB	WB	DB	WB	DB	WB	DB	WB	DB	WB		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	14.0	6.2	7.0	10.1	12.5	16.1	20.2	22.3	22.4	19.1	15.5	9.6	6.8
	DBStd	6.78	3.63	3.68	3.12	3.06	3.34	3.64	3.29	3.26	3.18	3.61	3.71	3.90
	HDD10.0	429	123	93	37	11	2	0	0	0	4	50	110	
	HDD18.3	1989	375	318	254	176	85	22	7	5	28	101	262	357
	CDD10.0	1899	7	9	41	86	191	306	382	384	272	174	38	11
	CDD18.3	417	0	0	0	1	15	78	131	131	51	12	0	0
	CDH23.3	3754	0	0	1	19	144	726	1229	1190	399	46	0	0
	CDH26.7	1356	0	0	0	1	27	256	483	480	106	4	0	0
(14) Wind	WSAvg	3.9	3.8	4.2	4.4	4.3	4.1	3.9	3.8	3.5	3.5	3.7	3.7	3.6
(15) Precipitation	PrecAvg	654	54	47	52	65	74	62	42	47	50	54	53	52
	PrecMax	920	146	95	118	171	180	265	185	151	151	162	146	168
	PrecMin	406	2	6	0	10	20	7	2	9	0	10	3	3
	PrecStd	122	31	25	27	35	34	40	31	29	35	39	28	34
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	15.8	18.3	22.4	25.9	29.4	34.4	35.4	36.9	31.7	27.0	20.4	16.4
		MCWB	12.1	11.9	13.1	16.4	18.3	21.1	20.8	21.2	19.6	18.0	15.2	12.9
	2%	DB	13.9	15.6	19.5	22.9	26.9	31.4	33.0	33.7	29.6	24.5	18.3	14.7
		MCWB	11.2	10.8	12.5	14.4	17.4	20.2	20.7	20.9	18.8	17.7	14.3	11.9
	5%	DB	12.8	13.9	17.5	20.6	24.8	29.6	31.3	31.3	27.6	22.6	16.7	13.5
		MCWB	10.5	10.1	11.4	13.5	16.9	19.5	20.3	20.6	18.2	16.8	13.6	11.2
	10%	DB	11.6	12.4	15.7	18.6	22.6	27.7	29.5	29.3	25.6	21.0	15.2	12.2
		MCWB	9.7	9.5	10.9	12.6	15.8	18.9	19.8	20.0	17.7	16.5	12.7	10.4
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	13.0	12.6	14.5	16.8	19.9	22.9	23.1	23.5	21.4	19.6	16.1	13.3
		MCDB	14.7	16.0	19.4	23.9	27.0	31.3	31.4	31.0	28.6	24.2	19.3	15.5
	2%	WB	11.7	11.5	13.1	15.4	18.5	21.5	22.1	22.4	20.4	18.7	15.0	12.4
		MCDB	13.5	14.5	17.5	21.4	24.9	29.4	29.9	29.6	26.4	22.3	17.5	14.2
	5%	WB	10.9	10.6	12.2	14.2	17.4	20.4	21.4	21.6	19.5	18.0	13.9	11.5
		MCDB	12.5	13.2	16.3	19.2	23.1	27.6	28.7	28.5	25.1	21.3	16.2	13.2
	10%	WB	9.8	9.8	11.4	13.2	16.4	19.5	20.6	20.8	18.6	17.1	12.9	10.6
		MCDB	11.3	12.0	14.9	17.4	21.4	26.1	27.5	27.5	23.8	20.3	14.8	12.2
(35) Mean Daily Temperature Range	5% DB	MDBR	6.6	7.8	9.1	9.6	10.0	10.9	11.2	11.2	10.7	8.9	7.2	6.8
		MCDBR	7.8	10.4	13.1	13.5	13.6	14.3	14.8	14.8	13.9	10.8	9.1	8.0
	5% WB	MCWBR	5.5	6.3	7.1	6.7	6.3	5.4	5.1	5.0	5.3	5.2	5.6	5.7
		MCWBR	6.8	8.9	10.3	11.3	11.7	12.6	12.4	12.4	11.3	8.5	7.7	7.1
(40) Clear-Sky Solar Irradiance	taub		0.321	0.336	0.374	0.398	0.407	0.418	0.410	0.410	0.379	0.365	0.339	0.319
		taud	2.476	2.412	2.310	2.249	2.281	2.284	2.306	2.317	2.388	2.432	2.460	2.480
	Ebn at Noon		809	855	860	866	866	855	857	844	846	809	776	774
		Edh at Noon	72	91	114	131	130	130	126	121	104	88	73	66
(44) All-Sky Solar Radiation	RadAvg		1.44	2.38	3.62	4.68	5.34	6.14	6.08	5.43	4.35	2.89	1.70	1.34
	RadStd		0.13	0.27	0.34	0.50	0.39	0.44	0.43	0.33	0.22	0.23	0.19	0.17

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (24 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page